4 Instruction Set of 8085

Opcode and Operands

Each instruction consists of two parts: opcode and operand. The first part is the opcode, which communicates the task to be done by the instruction; and the latter part, operand, provides the data to be operated on. Data may be 8/16 bit data, 8/16 bit address, register, or memory location. If the operand is a register, it means the data are present in that register.

Instruction Word Size

According to the word size of instruction, there are three types of instruction.

- 1 byte instruction (e.g., MOV A, B; ADD B; RAL)
- 2 byte instruction (e.g., MVI B, 05; IN 01)
- 3 byte instruction (e.g., LXI H, 2022; STA 2500)

Addressing Modes

Depending on the mode of addressing the data in an instruction, these are the following addressing modes.

- Register addressing mode
- Direct addressing mode
- Indirect addressing mode
- Immediate addressing mode
- Implied addressing mode

Register addressing mode: in this addressing mode, operand is in general purpose register(s) or accumulator. Examples are MOV A,C; ADD C.

Direct addressing mode: in this addressing mode, address of the operand is given in the instruction itself. Examples are LDA 2500 H; STA 2600 H.

Indirect addressing mode: in this addressing mode, address of the operand is specified by a register pair. Examples are LXI H, 2300 H.

Immediate addressing mode: in this addressing mode, operand is given in the instruction itself. Examples are MVI B, 06: ADI 07.

Implied addressing mode: in this addressing mode, opcode specifies the address of the operands. Examples are RAR, CMA, RAL, etc.

Classification of Instructions

Depending on the task performed by the instruction, they are classified as follows.

- Data transfer instructions
- Arithmetic instructions
- Logical instructions
- Branch instructions
- Stack, I/O, and machine control instructions

Data transfer instructions: in these instructions, data are transferred from source to destination. Source can be data, contents of register, or contents of memory location. Destination can be a register or memory location. They do not affect any flag.

Arithmetic instructions: these instructions perform arithmetic operations, such as addition, subtraction, increment, and decrement. They affect the flag register.

Logical instructions: these instructions perform logical operations, such as AND, OR, XOR, compare, rotate, and complement. They affect the flag register.

Branch instructions: in these instructions, program sequence changes to serve some other instructions between a program, conditionally, or unconditionally. It includes jump, call, and return instructions.

Stack, I/O, and machine control instructions: Stack instructions transfer data from register pair to stack memory and from stack memory to register pair. I/O instructions transfer 8 bit data to I/O ports. Machine instructions control the machine operation, such as interrupt, halt, etc.

Instruction	Description	IWS	Addressing Mode	No. of Machine Cycles	T-States	Flag Affected
MOV R_d, R_s R_d, R_s→ A, B, C, D, E, H, and L	Copy the data of source register R_s to the destination register R_d.	1 byte	Register	1 (OP)	4T	None
MOV M, R	Copy the data of register R to the given memory location.	1 byte	Register	2 (OP+ MW)	7T	None
MOV R, M	Copy the data of memory location to the register R.	1 byte	Indirect	2 (OP+ MR)	7T	None
MVI R, 8 bit data	Copy the 8 bit data immediately to the register R.	2 byte	Immediate	2 (OP+MR)	7T	None
MVI M, 8 bit data	Copy the 8 bit data to the memory location.	2 byte	Indirect	3 (OP+MR+MW)	10T	None
LXI R_p ,16 bit data R_p→ BC, DE, HL, and SP	Load the 16 bit data immediately in register pair R_p.	3 byte	Immediate	3 (OP+MR+MR)	10T	None
LDA 16 bit address	Load accumulator directly with the content of given 16 bit address.	3 byte	Direct	4 (OP+MR+MR+MR)	13T	None
STA 16 bit address	Store the content of accumulator at given 16 bit address.	3 byte	Direct	4 (OP+MR+MR +MW)	13T	None

Instruction	Description	IWS	Addressing Mode	No. of Machine Cycles	T-States	Flag Affected
LDAX R_p R_p →BC and DE	Load accumulator with contents of memory location whose address is stored in register R_p.	1 byte	Indirect	2 (OP+MR)	7T	None
STAX R_p R_p →BC and DE	Store the content of accumulator at memory location whose address is stored in register.	1 byte	Indirect	2 (OP+MW)	7T	None
LHLD 16 bit address	Load HL pair with contents of given address and next address. [L]←given address [H] ←next address.	3 byte	Direct	5(OP+MR+MR +MR+MR)	16T	None
SPHL	The content of HL pair is copied in stack pointer.	1 byte	Register	1 (OP)	6T	None
PCHL	The content of HL pair is copied in program counter.	1 byte	Register	1 (OP)	6T	None
XCHG	Exchange the content of HL pair with DE pair.	1 byte	Register	1 (OP)	4T	None

Instruction	Description	IWS	Addressing Mode	No. of Machine Cycles	T-States	Flag Affected
IN 8 bit port address	Copy the contents of I/O mapped device connected at given 8 bit port address directly to accumulator.	2 byte	Direct	3 (OP+MR+IOR)	10T	None
OUT 8 bit port address	Copy the contents of accumulator directly to I/O mapped I/O device connected at given 8 bit port address.	2 byte	Direct	3 (OP+MR+IOW)	10T	None

Table 4.1 Data Transfer Instructions

Note: no flag is affected by data transfer instruction

Note: OP: Op code Fetch, MW: Memory Write, MR: Memory Read, IOR: I/O Read, IOW: I/O Write

Instruction	Description	IWS	Addressing Mode	No. of Machine Cycles	T-States	Flag Affected
ADD R	Add the content of R to A and store the result in A.	1 byte	Register	1(OP)	4T	All
ADD M	Add the content of memory location, whose address is stored in HL pair to A and store result in A.	1 byte	Register	2 (OP+ MR)	7T	All
ADI 8 bit data	Add the 8 bit data to A and store the result in A.	2 byte	Immediate	2 (OP+ MR)	7T	All

Instruction	Description	IWS	Addressing Mode	No. of Machine Cycles	T-States	Flag Affected
ADC R R→ A, B, C, D, E, H, and L	Add the content of R along with carry flag to contents of A and store the result in A.	1 byte	Register	1(OP)	4T	All
ADC M	Add the content of M along with carry flag to the content of A and store the result in A.	1 byte	Indirect	2(OP+MR)	7T	All
ACI 8 bit data	Add the 8 bit data along with carry flag to the contents of A and store the result in A.	2 byte	Immediate	2(OP+MR)	7T	All
DAD R_p R_p→ BC, DE, HL, and SP	Add the content of register pair R_p to the contents of HL pair and store the result in HL pair.	1 byte	Register	3(OP+BI+BI)	10T	Only carry (CY) flag
Decimal adjust accumulator (DAA)	Binary contents of accumulator are changed to its two digit BCD equivalent.	1 byte	Implicit	1(OP)	4T	All
SUB R R→ A, B, C, D, E, H, and L	Subtract the contents of R from contents of A and store the result in A.	1 byte	Register	1(OP)	4T	All
SUB M M→ Memory address stored in HL pair	Subtract the contents of M from A and store the result in A.	1 byte	Indirect	2(OP+ MR)	7T	All

Instruction	Description	IWS	Addressing Mode	No. of Machine Cycles	T-States	Flag Affected
SUI 8 bit data	Subtract the 8 bit data immediately from the content of A and store the result in A.	2 byte	Immediate	2(OP+MR)	7T	All
SBB R R→ A, B, C, D, E, H, and L	Subtract the content of R along with carry flag from the content of A and store the result in A.	1 byte	Register	1(OP)	4T	All
SBB M M→ Memory address stored in HL pair	Subtract the content of M along with the carry flag from A and store the result in A.	1 byte	Indirect	2(OP+MR)	7T	All
SBI 8 bit data	Subtract the 8 bit data along with carry flag from the content of A and store the result in A.	2 byte	Immediate	2(OP+MR)	7T	All
INR R R→ A, B, C, D, E, H, and L	Increment the content of R by 1.	1 byte	Register	1(OP)	4T	All flags except CY flag
INR M M→ Memory address stored in HL pair	Increment the content of M by 1.	1 byte	Indirect	3(OP+MR+MW)	10T	All flags except CY flag
DCR R R→ A, B, C, D, E, H, and L	Decrement the content of R by 1.	1 byte	Register	1(OP)	4T	All flags except CY flag

Instruction	Description	IWS	Addressing Mode	No. of Machine Cycles	T-States	Flag Affected
DCR M M→ Memory address stored in HL pair	Decrement the content of M by 1.	1 byte	Indirect	3(OP+MR+MW)	10T	All flags except CY flag
INX R_p R_p→ BC, DE, HL, and SP	Increment the content of register pair R_P by 1.	1 byte	Register	1(OP)	6T	None
DCX R_P R_p→ BC, DE, HL, and SP	Decrement the content of register pair R_P by 1.	1 byte	Register	1(OP)	6T	None

Table 4.2 Arithmetic Instructions

Note: DAA instruction changes the binary values of the contents of the accumulator to its two digit equivalent BCD number. This instruction makes use of the status of CY and AC flags. It performs the operation as follows.

- If the value of the lower 4 bits (D_3–D_0) in the accumulator is greater than 9, or if AC flag is set, the instruction adds 0110 to the lower order bits.
- If a value of higher order 4 bits (D_7 – D_4) in the accumulator is greater than 9, or if CY flag is set, the instruction adds 0110 to higher order 4 bits.

Note: DAA is used for BCD addition only not for BCD subtraction.

Instruction	Description	IWS	Addressing Mode	No. of Machine Cycles	T-States	Flag Affected
ANA R R→ A, B, C, D, E, H, and L	Contents of R are ANDed with the contents of A and the result is stored in A.	1 byte	Register	1(OP)	4T	All flags. CY flag is reset and AC flag is set.
ANA M M→ Memory address stored in HL pair	Contents of M are ANDed with contents of A and the result is stored in A.	1 byte	Indirect	2(OP+MR)	7T	All flags. CY flag is reset and AC flag is set.

Instruction	Description	IWS	Addressing Mode	No. of Machine Cycles	T-States	Flag Affected
ANI 8 bit data	8 bit data are ANDed with contents of A and the result is stored in A.	2 byte	Immediate	2(OP+MR)	7T	All flags. CY flag is reset and AC flag is set.
ORA R R→ A, B, C, D, E, H, and L	Contents of R are ORed with contents of A and the result is stored in A.	1 byte	Register	1(OP)	4T	All flags. Both CY and AC flags are reset.
ORA M M→ Memory address stored in HL pair	Contents of M are ORed with contents of A and the result is stored in A.	1 byte	Indirect	2(OP+MR)	7T	All flags. Both CY and AC flags are reset.
ORI 8 bit data	8 bit data are ORed with contents of A and the result is stored in A.	2 byte	Immediate	2(OP+MR)	7T	All flags. Both CY and AC flags are reset.
XRA R R→ A, B, C, D, E, H, and L	Contents of R are EX-ORed with contents of A and the result is stored in A.	1 byte	Register	1(OP)	4T	All flags. Both CY and AC flags are reset.
XRA M	Contents of M are EX-ORed with the contents of A and the result is stored in A.	1 byte	Indirect	2(OP+MR)	7T	All flags. Both CY and AC flags are reset.
XRI 8 bit data	8 bit data are EX-ORed with the contents of A and the result is stored in A.	2 byte	Immediate	2(OP+MR)	7T	All flags. Both CY and AC flags are reset.

Instruction	Description	IWS	Addressing Mode	No. of Machine Cycles	T-States	Flag Affected
CMA	Complement the contents of accumulator.	1 byte	Implicit	1(OP)	4T	None
CMC	Complement the carry flag.	1 byte	None	1(OP)	4T	CY flag is complemented. No other flag is affected.
STC	Set the carry flag.	1 byte	None	1(OP)	4T	CY flag is set. No other flag is affected.
RLC	Rotate the contents of the accumulator to the left without carry (CY) flag.	1 byte	Implicit	1(OP)	4T	Only CY flag is affected.
RAL	Rotate the contents of the accumulator to left through carry (CY) flag.	1 byte	Implicit	1(OP)	4T	Only CY flag is affected.
RRC	Rotate the contents of the accumulator to the right without carry (CY) flag.	1 byte	Implicit	1(OP)	4T	Only CY flag is affected.
RAR	Rotate the contents of the accumulator to the right through carry (CY) flag.	1 byte	Implicit	1(OP)	4T	Only CY flag is affected.
CMP R	Compare the contents of R with contents of A.	1 byte	Register	1(OP)	4T	All

Instruction	Description	IWS	Addressing Mode	No. of Machine Cycles	T-States	Flag Affected
CMP M	Compare the contents of M with contents of A.	1 byte	Indirect	2(OP+MR)	7T	All
CPI 8 bit data	Compare the 8 bit data with the contents of A.	2 byte	Immediate	2(OP+MR)	7T	All

Table 4.3 Logical Instructions

Note: In CMP, R instruction comparison is made by subtracting the content of register R from content of the accumulator. The contents of A and R remain unaltered after comparison. The result of the operation is reflected by status of flag as shown below.

If A-R › 0 then S=0 ; Z=0 ; CY=0

If A-R = 0 then S=0 ; Z=1 ; CY=0

If A-R ‹ 0 then S=1 ; Z=0 ; CY=1

Note: In CMP, M instruction comparison is made by subtracting the content of M from the content of accumulator. The contents of A and M remain unaltered after comparison. The result of operation is reflected by the status of flag as shown below.

If A-M › 0 then S=0 ; Z=0 ; CY=0

If A-M = 0 then S=0 ; Z=1 ; CY=0

If A-M ‹ 0 then S=1 ; Z=0 ; CY=1

Note: In CPI, 8 bit data M instruction comparison are made by subtracting the 8 bit data from the content of the accumulator. The contents of A remain unaltered after comparison. The result of operation is reflected by the status of flag as shown below.

If A-(8-bit data) › 0 then S=0 ; Z=0 ; CY=0

If A-(8-bit data) = 0 then S=0 ; Z=1 ; CY=0

If A-(8-bit data) ‹ 0 then S=1 ; Z=0 ; CY=1

Instruction	Description	IWS	Addressing Mode	No. of Machine Cycles	T-States	Flag Affected
HLT	Halt the execution of program.	1 byte	No	Two or more machine cycles	5 or more T-states	None
NOP	No operation is performed.	1 byte	No	1(OP)	4T	None

Table 4.4 Machine Control Instructions

Note: When HLT statement is given, the microprocessor completes the execution of current instruction and halts further execution.

Note: NOP instruction is used to introduce a delay of 4 T-states between two successive instructions.

Jump Instructions	Instruction	Description	IWS	Addressing Mode	No. of Machine Cycles	T states	Flag Affected
Unconditional Jump Instruction	JMP 16 bit address	Jump execution of program immediately at the given 16 bit address	3 byte	Immediate	3(OP+MR +MR)	10T	None
Conditional Jump Instruction	JC 16 bit address	Jump execution of program immediately at the given 16-bit address, if CY = 1	3 byte	Immediate	3(OP+MR+MR), if condition is true. 2, if condition is false.	10T if condition is true. 7T if condition is false.	None
	JNC 16 bit address	Jump execution of program immediately at the given 16 bit address, if CY = 0	3 byte	Immediate	3(OP+MR+MR), if condition is true. 2, if condition is false.	10T if condition is true. 7T if condition is false.	None
	JZ 16 bit address	Jump execution of program immediately at the given 16 bit address, if Z = 1	3 byte	Immediate	3(OP+MR+MR), if the condition is true. 2, if condition is false.	10T if the condition is true. 7T if condition is false.	None
	JNZ 16 bit address	Jump execution of program immediately at the given 16 bit address, if Z = 0	3 byte	Immediate	3(OP+MR+MR), if the condition is true. 2, if condition is false.	10T, if the condition is true. 7T, if the condition is false.	None

Jump Instructions	Instruction	Description	IWS	Addressing Mode	No. of Machine Cycles	T states	Flag Affected
	JP 16 bit address	Jump execution of program immediately at the given 16 bit address, if S = 0	3 byte	Immediate	3(OP+MR+MR), if the condition is true. 2, if condition is false.	10T, if the condition is true. 7T, if condition is false.	None
	JM 16 bit address	Jump execution of program immediately at the given 16 bit address, if S = 1	3 byte	Immediate	3(OP+MR+MR), if condition is true. 2, if condition is false.	10T, if condition is true. 7T, if condition is false.	None
	JPE 16 bit address	Jump execution of program immediately at the given 16 bit address, if P = 1	3 byte	Immediate	3(OP+MR+MR), if the condition is true. 2, if condition is false	10T, if the condition is true. 7T, if condition is false.	None
	JPO 16 bit address	Jump execution of program immediately at the given 16 bit address, if P = 0	3 byte	Immediate	3(OP+MR+MR), if the condition is true. 2, if condition is false.	10T, if the condition is true. 7T, if condition is false.	None

Table 4.5 Jump Instructions

Stack and its related instruction: stack is a group of memory locations used by a microprocessor or programmer to store the data temporarily during the execution of a program. The stack can be used to store the contents of program counter (PC) by the microprocessor automatically when a subroutine is called. The stack may be used by the programmer for storing the content of register pairs of BC, DE, HL, and program status word (PSW). Program status word indicates the contents of the accumulator and flag register. Here, the accumulator is considered as a higher order and flag register is considered as a lower order register.

Instruction	Description	IWS	Addressing Mode	No. of Machine Cycles	T-States	Flag Affected
PUSH R_P R_P→ BC, DE, HL, and PSW	Store the content of register pair RP on the two top locations of stack.	1 byte	Register Indirect	3(OP +MW+MW)	12T	None
POP R_P R_P→ BC, DE, HL, and PSW	Retrieve the contents of two top locations of stack to register pair RP.	1 byte	Register Indirect	3(OP+MR+MR)	10T	None
XTHL	Exchange the contents of two top locations of stack with the contents of HL pair.	1 byte	Register Indirect	5(OP+MR+MR +MW+MW)	16T	None

Table 4.6

Note: PUSH instruction first decrements the SP by one, and then, copies the contents of higher order register of pair R_P on the location shown by stack pointer. Then, the stack pointer register is decremented by one and the content of the lower order register of pair R_P is copied on the location shown by stack pointer. Thus, stack pointer is decremented by two after the execution of PUSH instruction.

Note: POP instruction first copies the contents of memory location indicated by stack pointer (SP) to lower order register of pair R_P, and then, increment stack pointer by one. It again copies the contents of memory location indicated by SP to higher order register of pair R_P, and then, increment stack pointer by one. Thus, the stack pointer is incremented by two after the execution of each POP instruction.

Note: after the execution of XTHL instruction, the contents of L are exchanged with the location indicated by SP and contents of H are exchanged with the memory location (SP+1). However, the content of SP remains unaltered.

Call instructions: these are used to call a subroutine during the execution of the main program. A subroutine is a set of instructions used to perform a task repeatedly.

Jump Instructions	Instruction	Description	IWS	Addressing Mode	No. of Machine Cycles	T-states	Flag Affected
Unconditional Call Instruction	CALL 16 bit address	Call immediately the subroutine stored at the given 16 bit address.	3 byte	Immediate or register indirect	5(OP+MR +MR+MW+MW)	18T	None
Conditional Call Instruction	CC 16 bit address	Call immediately the subroutine stored at the given 16 bit address, if CY = 1.	3 byte	Immediate or register indirect	5(OP+MR+MR +MW+MW), if condition is true. 2, if condition is false.	18T, if condition is true. 9T, if condition is false.	None
	CNC 16 bit address	Call immediately the subroutine stored at the given 16 bit address, if CY = 0.	3 byte	Immediate or register indirect	5(OP+MR+MR +MW+MW), if condition is true. 2, if condition is false.	18T, if condition is true. 9T, if condition is false	None
	CZ 16 bit address	Call immediately the subroutine stored at the given 16 bit address, if Z = 1.	3 byte	Immediate or register indirect	5(OP+MR+MR +MW+MW), if condition is true. 2, if condition is false.	18T, if condition is true. 9T, if condition is false.	None
	CNZ 16 bit address	Call immediately the subroutine stored at the given 16 bit address, if Z = 0.	3 byte	Immediate or register indirect	5(OP+MR+ MR+MW+MW), if condition is true. 2, if condition is false	18T, if condition is true. 9T, if condition is false.	None
	CP 16 bit address	Call immediately the subroutine stored at the given 16 bit address, if S = 0.	3 byte	Immediate or register indirect	5(OP+MR+MR +MW+MW), if condition is true. 2, if condition is false.	18T, if condition is true. 9T, if condition is false.	None

Jump Instructions	Instruction	Description	IWS	Addressing Mode	No. of Machine Cycles	T-states	Flag Affected
	CM 16 bit address	Call immediately the subroutine stored at the given 16-bit address, if S = 1.	3 byte	Immediate or register indirect	5(OP+MR+MR +MW+MW), if condition is true. 2, if condition is false.	18T, if condition is true. 9T, if condition is false.	None
	CPE 16 bit address	Call immediately the subroutine stored at the given 16-bit address, if P = 1.	3 byte	Immediate or register indirect	5(OP+MR+MR +MW+MW), if condition is true. 2, if condition is false	18T, if condition is true. 9T, if condition is false.	None
	CPO 16 bit address	Call immediately the subroutine stored at the given 16 bit address, if P = 0	3 byte	Immediate or register indirect	5(OP+MR+MR +MW +MW), if condition is true. 2, if condition is false.	18T, if condition is true. 9T, if condition is false.	None

Table 4.7

Practice Exercise

Level I

1. The total number of memory accesses involved (inclusive of the opcode fetch) when an 8085 processor executes the instruction LDA 2003 is:
(A) 1
(B) 2
(C) 3
(D) 4

2. In an 8085 microprocessor, the instruction CMP B has been executed while the content of the accumulator is less than that of register B. As a result:
(A) carry flag will be set but zero flag will be reset.
(B) carry flag will be reset but zero flag will be set.
(C) both carry flag and zero flag will be re reset.
(D) both carry flag and zero flag will be reset.

3. The number of memory cycles required to execute the following 8085 instructions:
I. LDA 3000 H
II. LXI D, F0F1 H
Would be:
(A) 2 for (I) and 2 for (II)
(B) 4 for (I) and 3 for (II)
(C) 3 for (I) and 3 for (II)
(D) 3 for (I) and 4 for (II)

4. Consider the sequence of 8085 instructions given below.

LXI H, 9258

MOV A, M

CMA

MOV M, A

Which one of the following is performed by this sequence?

(A) The contents of location 9258 are moved to the accumulator.

(B) The contents of location 9258 are compared with the contents of the

(C) The contents of location 8529 are complemented and stored in location 8529.

(D) The contents of location 5892 are complemented and stored in location 5892.

5. If the HLT instruction of an 8085 microprocessor is executed:

(A) the microprocessor is disconnected from the system till rest is pressed

(B) the microprocessor enters an HAL state and the uses are tri-stated

(C) the microprocessor halts the execution of the program and returns to monitor

(D) the microprocessors reload the program from the locations 0024 and 0025 H

6. The contents of the accumulator in an 8085 microprocessor are altered after the execution of the instruction:

(A) CMPC

(B) CPI 3A

(C) ANI 5 C

(D) ORA A

7. In an 8085 microprocessor, after the execution of XRA A instruction:

(A) the carry flag is set.

(B) the accumulator contains FF_H.

(C) the zero flag is set.

(D) the accumulator contents are shifted to the left by one bit.

8. The 8085 assembly language instruction that stored the contents of H and L registers into the memory location $205I_H$ and 2050_H, respectively, is:

(A) SPHL 2050_H

(B) SPHL 2051_H

(C) SHLD 2050_H

(D) STAX 2050_H

9. The 8085 microprocessor, CY flag may be set by the instruction:

(A) SUB

(B) INX

(C) CMA

(D) ANA

10. For an 8085 microprocessor, the Stack Pointer (SP) and Program Counter (PC) registers contain the numbers 2400_H and $F000_H$ in Hex, respectively. The contents of the register after the execution of the instruction CALL $E000_H$ would be:

(A) PC: $F003_H$ SP: 2400_H

(B) PC: $E000_H$ SP: 2400_H

(C) PC: $E000_H$ SP: 2401_H

(D) PC: $E000_H$ SP: $23FE_H$

11. Find the correct match among the following pair in the context of an 8085 microprocessor.

List-I	**List-II**
P. DAA	I. Program control instruction
Q. LXI	II. Data movement instruction
R. RST	III. Interrupt instruction
S. JMP	IV. Arithmetic instruction

(A) P-I, Q-II, R-III, S-IV

(B) P-IV, Q-II, E-III, S-I

(C) P-IV, Q-II, R-III, S-I

(D) P-II, Q-IV, R-III, S-I

12. A memory mapped I/O device has as address of 00F0H. Which of the following 8085 instructions outputs the content of the accumulator to the I/O device?

(A) LXIH, 00F0H
MOV M, A

(B) LXIH, OOFOH
OUT M

(C) LXIH, OOFOH
OUT F0H

(D) LXIH, 00F0H
MOV A, M

13. In an 8085 A microprocessor based system, it is desired to increment the contents of memory location, whose address is available in (D, E) register pair and store the result in same location. The sequence of the instruction is:

(A) XCHG
INR M

(B) XCHG
INX H

(C) INX d
XCHG

(D) INR m
XCHG

14. An 8085 assembly language program is given below.

Line I: MVI A, B5H
II: MVI B, 0EH
III: XRI 69H
IV: ADD B
V: ANI 9BH
VI: CPI 9FH
VII: STA 3010H
VIII: HLT

The contents of the accumulator just after the execution of the ADD instruction in line IV will be:

(A) C3H
(B) EAH
(C) DCH
(D) 69H

15. In an 8085 microprocessor, the contents of the accumulator, after the following instructions are executed will become:

XRA A
MVI B, F0H

(A) 01 H
(B) 0F H
(C) F0H
(D) 10 H

16. The following is an assembly language program for 8085 microprocessors.

Address	Instruction Code	Mnemonic
1000H	3E06	MVI A, 06H
1002H	C670	ADI 70H
1004H	32 0710	STA 1007H
1007H	AF	XRA A
1008H	76	HLT

When this program halts, the accumulator contains:

(A) 00H
(B) 06H
(C) 70H
(D) 76H

17. For an 8085 assembly language program given below, the content of the accumulator after the execution of the program is:

3000 MIV A, 45 H
3002 MOV B, A
3003 STC
3004 CMC
3005 RAR
3006 XRA B

(A) 00H
(B) 45H
(C) 67H
(D) E7H

18. When a 'CALL Addr' instruction is executed, the CPU carries out the following sequential operations internally.

Note:

(R) means content of register R

((R)) means content of memory location pointed to by R

PC means Program Counter

SP means Stack point

(A) (SP) incremented
(PC) ← Addr
((SP)) ← (PC)

(B) (PC) ← Addr
((SP)) ← (PC)
(SP) incremented

(C) (PC) ← Addr
(SP) incremented
((SP)) ← (PC)

(D) ((SP)) ← (PC)
(SP) incremented
(PC) ← Addr

19. In an 8085 processor, the main program cells subroutine SUB1 is given below. When the program returns to the main program after executing SUB1, the value in the accumulator is:

Address	Opcode		Mnemonic
2000	3E 00	SUB 1:	MVI A, 00H
2002	CD 05 20		CALL SUB 2
2005	3C	SUB 2:	INR A
2006	C9		RET

Choose the correct option.

(A) 00H
(B) 01 H
(C) 02 H
(D) 03 H

20. The subroutine SBX given below is executed by an 8085 processor. The value in the accumulator immediately after the execution of the subroutine will be:

```
SBX:    MVI A, 99 H
        ADI  11H
        MOV C, A
        RET
```

Choose the correct option.

(A) 00 H
(B) 11 H
(C) 99 H
(D) AA H

21. An 8085 assembly language program is given below. Assume that the carry flag is initially unset. The contents of the accumulator after the execution of the program is:

```
MVI A, 07H
RLC
MOV B, A
RLC
RLC
ADD B
RRC
```

Choose the correct option.

(A) 8CH
(B) 64 H
(C) 23 H
(D) 15

22. A portion of the main program to call a subroutine SUB in an 8085 environment is given below.

```
        :
        :
        LXI D, DISP
LP:     CALL SUB
        :
        :
```

It is desired that control be returned to LP + DISP + 3 when RET instruction is executed in the subroutine. The set of instructions that precede the RET instruction in the subroutine are:

(A) POP D
DAD H
PUSH D

(B) POP H
DAD D
INX H
INX H
INX H
PUSH H

(C) POP H
DAD D
PUSH H

(D) XTHL
INX D
INX D
INX D
XTHL

23. The contents (in hexadecimal) of some of the memory location in an 8085A based system are given below.

Address	Contents
• •	• •
26FF	00
26FF	01
2700	02
2701	23
2702	04
• •	• •

The contents of stack pointer (SP), program counter (PC), and (HL) are 2700H, 2100H, and 0000H, respectively. When the following sequence of instructions is executed:

2100 H: DAD SP
2101 H: PCHL

what will be the contents of (SP) and (PC) at the end of the execution?

(A) (PC) = 2102H, (SP) = 2700H
(B) (PC) = 2700H, (SP) = 2700H
(C) (PC) = 2800H, (SP) = 26FEH
(D) (PC) = 2A02H, (SP) = 2702H

24. An input device is interfaced with Intel 8085 A microprocessor as memory mapped I/O. The address of the device is 2500H. In order to input data from the device to the accumulator, the sequence of instructions will be:

(A) LXI H, 2500H
MOV A, M

(B) LXI H, 2500H
MOV M, A

(C) LHLD 2500H
MOV A, M

(D) LHLD 2500H
MOV M, A

25. Which one of the following addressing techniques is not used in an 8085 microprocessor?

(A) Register
(B) Immediate
(C) Register indirect
(D) Relative

26. When the operand required for an instruction is stored inside the processor, then what is this addressing mode called?

(A) Direct
(B) Register
(C) Implicit
(D) Immediate

27. Which one of the following instructions is a 3-byte instruction?

(A) MVI A
(B) LDAX B
(C) JMP 2050
(D) MOV A, M

28. Match List I (type of instruction) with List II (instruction) and select the correct answer using the codes given below the lists.

List I

a. One byte instruction
b. Two byte instruction
c. Three byte instruction
d. Register indirect addressing

List II

I. MVI A, data
II. MOV A, B
III. MOV A, M
IV. JMP addr

Codes:

	a	b	c	d
(A)	I	II	IV	III
(B)	II	I	IV	III
(C)	II	I	III	IV
(D)	I	II	III	IV

29. In the 8085 microprocessor system, the direct addressing instruction is:

(A) MOV A, B
(B) MVI B, 0AH
(C) MOV C, M
(D) STA addr

30. Match List I (instructions) with List II (type of addressing) with reference to an 8085 microprocessor and select the correct answer using the codes given below the lists.

List I

a. MOV A, M
b. LXI H, E400H
c. LDA F1CDH
d. CMC

List II

I. Direct addressing
II. Register addressing
III. Implicit addressing
IV. Register indirect addressing
V. Immediate addressing

Codes:

	a	b	c	d
(A)	V	IV	I	III
(B)	IV	V	III	I
(C)	V	IV	II	III
(D)	IV	V	I	III

31. For Intel 8085, match List I (addressing mode) with List II (instruction) and select the correct answer using the codes given below the lists.

List I

a. Implicit addressing
b. Register indirect
c. Immediate
d. Direct addressing

List II

I. JMP 3FA0H
II. MOV A, M
III. LDA 03FCH
IV. RAL

Codes:

	a	b	c	d
(A)	IV	I	II	III
(B)	IV	II	I	III
(C)	III	II	I	IV
(D)	III	I	II	IV

32. For the purpose of data processing, an efficient assembly language programmer makes use of the general purpose register rather than memory. The reason is:

(A) the set of instructions for data processing with memory is limited.
(B) data processing becomes easier when registers are used.
(C) more memory related instructions are required in the program for data processing.
(D) data processing with registers takes fewer cycles than that with memory.

33. Match List I (character used in programming) with List II (assembly language) and select the correct answer using the codes given below the lists.

List I

a. Blank
b. Colon
c. Semicolon
d. Comma

List II

I. Beginning of comment field
II. End of label field
III. Separator for operand
IV. Separator for op field and operand

Codes:

	a	b	c	d
(A)	I	III	IV	II
(B)	IV	II	I	III
(C)	I	II	IV	III
(D)	IV	III	I	II

34. A sequence of instruction is executed as follows.

Begin
Decode the instruction fetched.
If operand in memory then fetch operand.
Execute instructions.
End

This sequence of instructions has performed which of the following?

(A) Fetch cycle
(B) Both fetch and execute cycles
(C) Execute and decode cycles
(D) Execute cycle

35. Match List I (instruction) with List II (operation) for Intel 8085 and select the correct answer using the codes given below the lists.

List I

a. PCHL
b. SPHL
c. XTHL
d. XCHG

List II

I. Exchange the top of the stack with the contents of HL pair.
II. Exchange the contents of HL with those of DE pair.
III. Transfer the contents of HL to the stack pointer.
IV. Transfer the contents of HL to the program counter.

Codes:

	a	b	c	d
(A)	III	IV	I	II
(B)	III	IV	II	I
(C)	IV	III	II	I
(D)	IV	III	I	II

36. An example of an 8285 instruction that uses direct addressing is:

(A) RLC
(B) STA
(C) RRC
(D) CMA

37. Match List I (instructions) with List II (application) and select the correct answer using the codes given below the lists.

List I

a. SIM
b. DAD
c. DAA
d. SPHL

List II

I. 16 bit addition
II. Initialising the stack pointer
III. Serial output data
IV. Checking the current interrupt mask settings
V. CD addition

Codes:

	a	b	c	d
(A)	V	IV	II	I
(B)	III	I	V	II
(C)	V	I	II	IV
(D)	III	IV	V	I

38. Consider the following set of instructions.
STC
CMC
MOV A, B
RAL
MOV B, A
This set of instructions:
(A) doubles the number in register by B.
(B) divides the number in register by 2.
(C) multiplies B by A.
(D) adds A and B.

39. An RRC instruction in an 8085 microprocessor instruction will affect:
(A) CY, S, flag
(B) CY flag
(C) S flag
(D) Z flag

40. In an 8085 microprocessor, the following sequence of instructions is executed.
STC
CMC
MOV A, B
RAL
MOV B, A
After the last instruction, the output will:
(A) rotate the contents of the accumulator and store it in B.
(B) get the contents of B register into the accumulator and rotate it to the left by one bit.
(C) double contents of B register.
(D) manipulate carry in A and B.

41. Consider the following assembly language program.
MVI A, 30 H
ADI 30 H
XRA A
POP H
After the execution of the above program, the contents of the accumulator will be:
(A) 30 H
(B) 60 H
(C) 00 H
(D) Contents of stack

42. In an 8085 microprocessor, whenever the POP H instruction is executed:
(A) data bytes in the HL pair are stored in the stack.
(B) two data bytes at the top of the stack are transferred to the HL pair.
(C) two data bytes at the top of the stack are transferred to the PC.
(D) two data bytes from the HL registers that were previously stored on the stack are transferred back to the HL pair.

43. Which one of the following is correct in connection with the commands of CMP and CPI?
(A) CMP takes less number of m/c cycles compared to CPI.
(B) CMP takes more number of m/c cycles compared to CPI.
(C) Both CMP and CPI take the same number of m/c cycles.
(D) CMP takes either less or the same number of m/c cycles as compared to CPI.

44. When RET instruction is executed by any subroutine, then:
(A) the top of the stack will be popped out and assigned to the PC.

(B) without any operation, the calling program would resume from instruction immediately following the call instruction.
(C) the PC will be incremented after the execution of the instruction.
(D) without any operation, the calling program would resume from instruction immediately following the call instruction, and also the PC will be incremented after the execution of the instruction.

45. Consider the following set of 8085 instructions used to read a byte of data from the output of an ADC. The byte represents digital equivalent of analog input voltage V_{in} applied to ADC when $\overline{RD}$ is asserted.

```
ADC      :  EQU   30H
GETADC   :  IN    ADC
            RET
```

When RET is executed:

I. opcode of 1N is fetched.
II. port address 30 H is decoded.
III. opcode of 1N is decoded.
IV. I/O read operation is performed.

The correct sequence of these operations is:

(A) III, I, IV, and II
(B) I, III, II, and IV
(C) I, III, IV, and II
(D) III, I, II, and IV

46. Which logical operation is performed by the ALU of an 8085 microprocessor to complement a number?

(A) AND
(B) NOT
(C) OR
(D) Exclusive OR

47. Which one of the following functions is performed by an 8085 instruction MOV H, C?

(A) Moves the contents of H register to C register
(B) Moves the contents of C register to H register
(C) Moves the contents of C register to HL pair
(D) Moves the contents of HL pair to C register

48. After an arithmetic operation, the flag register of an 8085 microprocessor has the following look.

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
1	0	X	1	X	0	X	1

The arithmetic operation results in:

(A) a carry and an odd parity number having 1 as the MSB.
(B) zero and the auxiliary carry flag are being set.
(C) a number with even parity and 1 as the MSB.
(D) a number with odd parity and 0 and the MSB.

49. Examine the following instruction to be executed by an 8085 microprocessor. The input port has an address of 01H and has a data 05H to input:

```
1N   01
ANI  80
```

After the execution of the two instructions, the following flag portions may occur.

I. Zero flag is set.
II. Zero flag is reset.
III. Carry flag is cleared.
IV. Auxiliary carry flag is set.

Select the correct answer using the codes given below.

(A) I and III
(B) II, III, and IV
(C) III and IV
(D) I, III, and IV

50. How many and which type of machine cycles are needed to execute PUSH PSW by an Intel 8085 A microprocessor?

(A) 2, fetch and memory write
(B) 3, fetch and 2 memory write
(C) 3, fetch and 2 memory read
(D) 3, fetch, memory read, and memory write

Level II

1. If the accumulator of an Intel 8085 A microprocessor contains 37 H and the

previous operations have set the carry flag, the instruction ACI 56 H will result in:
(A) 8 E_H
(B) 94_H
(C) $7E_H$
(D) 84_H

2. Which one of the following Intel 8085 assembly language instructions does not affect the contents of the accumulator?
(A) CM A
(B) CMP B
(C) DA A
(D) ADD B

3. The contents of memory locations 2000 H, 2001 H and 2002 H are AA_H, BB_H, and CC_H, respectively. What are the contents of H and L registers after executing the following instructions in sequence?
LXI H, 2001_H
LHLD 2001_H
Select the correct answer using the codes given below.
(A) Contents of H and L registers are 20_H and 01_H, respectively.
(B) Contents of H and L registers are AA_H and BB_H, respectively.
(C) $Cont_H$ H, respectively.
(D) Contents of H and L registers are CC_H and BB_H, respectively.

4. In the 8085, the DAA instructions are used for:
(A) direct address accumulator
(B) double add accumulator
(C) decimal adjust accumulator
(D) direct access accumulator

5. Consider the following statements about register indirect addressing.
I. It helps in writing code that executes faster.
II. It helps in writing compact code.
III. It allows reuse of memory CPU data transfer instructions.
IV. It is essential for stack operations.
Which of the statements given above are correct?
(A) I, III, and IV
(B) I, II, and IV
(C) II, III, and IV
(D) I, II, and III

6. Consider the following 8085 instructions.
ANA A, ORAA, XRAA, SUBA, CMPA.
Now, consider the following statements.
I. All are arithmetic and logic instructions.
II. All cause the accumulator to be cleared irrespective of its original contents.
III. All reset the carry flag.
IV. All of them are 1-byte instructions.
Which of the statements given above is/are correct?
(A) I, II, III, and IV
(B) II only
(C) I, II, and IV
(D) I, III, and IV

7. INR instructions of 8085 do not affect carry flag. Which of the following is correct about INR instruction?
(A) Overflow cannot be detected.
(B) Overflow can be detected.
(C) If a program requires overflow to be detected, ADD instructions should be used instead of INR.
(D) It can be used to increase the contents of the BC register pair.

8. What are the number of machine cycles n, and the type of machine cycles carried out for PUSH B?
(A) n = 2, fetch, and memory write
(B) n = 3, fetch, and two memory write
(C) n = 3, fetch, memory write, and read
(D) n = 3, fetch, and two memory read

9. Consider the following instructions of 8085μP.
I. MOV M, A
II. ADC
III. MVI A, FF
IV. CMP M
Which of these causes a change in the status of flag(s)?
(A) I and II
(B) I, II, and III

(C) III and IV
(D) II and IV

10. **Assertion (A):** A subroutine is a program written separately from the main program to perform a function that occurs repeatedly in the main program.
Reason (R): A subroutine can be called by a CALL instruction.
(A) Both A and R are individually true and R is the correct explanation of A.
(B) Both A and R are individually true but R is not the correct explanation of A.
(C) A is true but R is false.
(D) A is false but R is true.

11. Consider the execution of the following segment of instructions in sequence.
MVI A, 91H
XRI 91H
Which of the following is correct?
(A) The content of the accumulator is 00H. Carry, auxiliary carry, and zero flag are set to 0, 1, and 0, respectively
(B) The content of the accumulator is 91H. Carry, auxiliary carry, and zero flag are set to 0, 0, and 1, respectively
(C) The content of the accumulator is 00H. Carry, auxiliary carry, and zero flag are set to 0, 0, and 1, respectively
(D) The content of the accumulator is 91H. Carry, auxiliary carry, and zero flag are set to 0, 1, and 0, respectively

12. If the HLT instruction of an Intel 8085A microprocessor is executed:
(A) the microprocessor is disconnected from the system till the RESET is pressed.
(B) the microprocessor halts the execution of the program and returns to the monitor.
(C) the microprocessor enters a HALT state and the buses and tri-stated.
(D) the microprocessor reloads the program counter from the locations 0024 H and 0025 H.

13. Match List I (feature of instruction) with List II (instruction) and select the correct answer using the codes given below the lists.
List I
a. Maskable interrupt
b. Signal
c. Instruction
d. Memory location 002C H
List II
I. RST 5.5
II. XTHL
III. SID
IV. RST 6.5
Codes:

	a	b	c	d
(A)	IV	I	II	III
(B)	II	III	IV	I
(C)	IV	III	II	I
(D)	II	I	IV	III

14. What is the number of machine cycles in the instruction LDA 2000H that consists of 13 T-states?
(A) 2
(B) 3
(C) 4
(D) 5

15. Which of the following is the addressing mode used in the instruction JMP F347 H in case of an Intel 8085 A microprocessor?
(A) Direct
(B) Register indirect
(C) Implicit
(D) Immediate

16. Which is one of the indirect addressing modes in the following instructions?
(A) LXI H 2050 H
(B) MOV A, B
(C) LDAX B
(D) LDA 2050 H

17. Carry flag is not affected after the execution of:
(A) ADD B
(B) SBB B
(C) INR B
(D) ORA B

18. Match List I with List II and select the correct answer using the codes given below the lists.

List I

a. Immediate addressing
b. Implied addressing
c. Register addressing
d. Direct addressing

List II

I. LDA 30 FF
II. MOV A, B
III. LXI H, 2050
IV. RRC

Codes:

	a	b	c	d
(A)	III	IV	II	I
(B)	II	I	III	IV
(C)	III	I	II	IV
(D)	II	IV	III	I

19. If 8085 adds 87 H and 79 H, then:
(A) both carry and zero flags will be set to 0.
(B) carry flag will be set to 0, zero flag to 1.
(C) carry flag will be set to 1, zero flag to 0.
(D) both carry and zero flags will be set to 1.

20. On the 8085, which of the following machine cycles are not used in the CALL instructions?
I. Instruction fetch
II. I/O
III. Memory read
IV. Memory write
Select the correct answer using the code(s) given below.
(A) II only
(B) I and IV
(C) II, III, and IV
(D) None of the above

21. The contents of the accumulator in an 8085 microprocessor are altered after the execution of the instruction?
(A) CMP C
(B) CPI 3A
(C) ANI 5C
(D) ORAA

22. In an 8085 microprocessor, after the execution of XRA A instruction:
(A) the carry flag is set.
(B) the accumulator contains FF_H.
(C) the content of accumulator is shifted by one.
(D) the zero flag is set.

23. Consider the following instructions of 8085 μP.
I. MOV M, A
II. ADD C
III. MVI A, FF
IV. CMP M
Which of these causes change in the status of flag(s)?
(A) I and II
(B) I, II, and III
(C) III and IV
(D) II and IV

24. Which of the following addressing techniques is not used in a 8085 microprocessor?
(A) Register
(B) Immediate
(C) Register indirect
(D) Relative

25. When the operand required for instruction is stored inside the processor, then what is this addressing mode called?
(A) Direct
(B) Register
(C) Implicit
(D) Immediate

26. What is the correct 8085 assembly language instruction that stores the contents of L and H registers in the memory locations 1080 H and 1081 H respectively?
(A) SPHL 1080 H
(B) SHLD 1080 H
(C) STAX 1080 H
(D) SPHL 1081 H

27. An 8085 microprocessor is executing the program as follows.

```
MVI A, 20 H
MVI H, 10
BACK : NOP
ADD B
       RLC
```

JNC BACK
HLT

How many times the instruction NOP will be executed?
(A) 4
(B) 3
(C) 2
(D) 1

28. The stack pointer of an 8085 microprocessor is ABCD H. At the end of the execution of the sequence of instructions, what will be the content of the stack pointer?

PUSH	PSW
XTHL	
PUSH	D
JMP	FC70 H

(A) ABCB H
(B) ABC9 H
(C) ABCA H
(D) ABC8 H

29. For which of the following, the instruction XRA A in an 8085 microprocessor can be used?
(A) Set the carry flag
(B) Set the zero flag
(C) Reset the carry flag and clear the accumulator
(D) Transfer FFH to the accumulator

30. An output device is interfaced with an 8085 microprocessor as memory mapped I/O. The address of the devices is 1000H. In order to output the data from the accumulator to the device, what will be the sequence of instructions?
(A) LXI H, 1000H
MOV A, M
(B) LXI H, 1000H
MOV M, A
(C) LHLD 1000H
MOV A, M
(D) LHLD 1000H
MOV M, A

31. Which stack is used in an 8085 microprocessor?
(A) FIFO
(B) FILO
(C) LIFO
(D) LILO

32. In an instruction of an 8085 microprocessor, how many bytes are present?
(A) One or two
(B) One, two, or three
(C) One only
(D) Two or three

33. Match List I (instruction code) with List II (addressing mode) and select the correct answer using the codes given below the lists.

List I
a. JMP 2021
b. LDAX B
c. IN 10H
d. RLC

List II
I. Direct addressing
II. Immediate addressing
III. Register indirect addressing
IV. Implicit addressing

Codes:

	a	b	c	d
(A)	IV	III	I	II
(B)	II	III	I	IV
(C)	IV	I	III	II
(D)	II	I	III	IV

34. Which one of the following addressing modes is used in the instruction PUSH B?
(A) Direct
(B) Register
(C) Register indirect
(D) Immediate

35. Consider the following instruction of an 8085 microprocessor.
I. MOV B, C
II. STA address
III. ORI byte

The correct sequence in the decreasing order of their respective memory space requirement is:
(A) III, II, and I
(B) I, III, and II
(C) I, II, and III
(D) II, III, and I

36. If the CALL instruction of 8085 in the main program is conditional, then RETURN instruction in the subroutine can be:
(A) conditional
(B) conditional unconditional
(C) determined by LDA instruction
(D) unconditional

37. In the following 8085 programs, how many times (decimal) is the DCR C executed?

```
        MVI C, 78 H
LOOP:   DCR C
        JNZ Loop
        HLT
```

(A) 119
(B) 120
(C) 78
(D) 77

38. A single instruction to clear the lower 4 bits of the accumulator in an 8085 assembly language is:
(A) XRI 0FH
(B) ANI F0H
(C) XRI F0H
(D) ANI 0FH

39. Which of the following instructions copies a byte of data into the accumulator from the memory address given in the instruction?
(A) LDA address
(B) LDAX B
(C) LHLD address
(D) STA address

40.
```
LXI     SP,     7FFFH
MVI     A, 25 H
XRI     02H
PUSH    H
MOV     A, L
ORI     10 H
HLT
```

What are the contents of A, H, L, SP, and PSW registers after executing the above set of instructions? Assume undefined flags always remain clear.
(A) 10H, 25H, 00H, 7FFFH, 00H, respectively
(B) 14H, 27H, 04H, 7FFFH, 04H, respectively
(C) 14H, 25H, 00H, 7FFFH, 04H, respectively
(D) 10H, 27H, 04H, 7FFFH, 00H, respectively

41.
```
MVI     A, AAH
ORI     FFH
RRC
CMC
INR     A
```

What are the contents of A and PSW registers, after executing the above set of instructions in sequence?
(A) AAH and 00H
(B) FFH and 66H
(C) 00H and 54H
(D) 00H and 00H

42. The field, which is never present in an assembly language statement, is:
(A) opcode
(B) operand
(C) continue
(D) comment

43. The opcode for the instruction 'add immediately to accumulator with carry' in the 8085 microprocessor is:
(A) ADI
(B) ACI
(C) ADC
(D) ADD

44. In an 8085 microprocessor, how many interrupts are maskable?
(A) Two
(B) Three
(C) Four
(D) Five

45. The _____ instruction allows control to transfer to a subroutine, which when complete, issues the _____ instruction to return control to the main program.
(A) CALL, JMP
(B) CALL, RET
(C) CALL, RSTI
(D) EI, DI

46. A single instruction to clear the lower 4 bits of the accumulator in an 8085 microprocessor is:
(A) XRI 0FH
(B) ANI F0H
(C) XRI F0H
(D) ANI 0FH

47. The addressing mode used in the instruction STAX B is:
(A) direct
(B) immediate
(C) implicit
(D) indirect

48. Which of the following statements is not correct?
(A) CMP A is a single byte instruction and CMA is not an instruction.
(B) The instruction SUB A sets the zero flag.
(C) Bus is a group of wires.
(D) Instruction INR C does not affect carry flag.

49. To execute the following instructions:
(i) LDA 00 H
(ii) LXI H, 2100 H
By an Intel 8085 microprocessor, the numbers of memory cycles required are:
(A) 2 for (i) and 2 for (ii)
(B) 4 for (i) and 2 for (ii)
(C) 3 for (i) and 3 for (ii)
(D) 4 for (i) and 3 for (ii)

50. In an 8085 microprocessor, during PUSH PSW operation, stack pointer is:
(A) decremented by one
(B) decremented by two
(C) incremented by one
(D) incremented by two

Answer Key

Level I

1. (D)	**2.** (A)	**3.** (B)	**4.** (A)	**5.** (B)	**6.** (C)	**7.** (C)	**8.** (C)	**9.** (A)	**10.** (D)
11. (B)	**12.** (A)	**13.** (A)	**14.** (B)	**15.** (D)	**16.** (D)	**17.** (C)	**18.** (D)	**19.** (C)	**20.** (D)
21. (C)	**22.** (C)	**23.** (B)	**24.** (A)	**25.** (D)	**26.** (B)	**27.** (C)	**28.** (B)	**29.** (D)	**30.** (D)
31. (B)	**32.** (D)	**33.** (B)	**34.** (B)	**35.** (D)	**36.** (D)	**37.** (B)	**38.** (A)	**39.** (B)	**40.** (C)
41. (C)	**42.** (B)	**43.** (D)	**44.** (A)	**45.** (B)	**46.** (B)	**47.** (B)	**48.** (A)	**49.** (D)	**50.** (B)

Level II

1. (A)	**2.** (B)	**3.** (D)	**4.** (C)	**5.** (C)	**6.** (C)	**7.** (A)	**8.** (B)	**9.** (D)	**10.** (B)
11. (C)	**12.** (C)	**13.** (C)	**14.** (C)	**15.** (D)	**16.** (C)	**17.** (C)	**18.** (A)	**19.** (D)	**20.** (A)
21. (A)	**22.** (D)	**23.** (D)	**24.** (D)	**25.** (B)	**26.** (B)	**27.** (B)	**28.** (C)	**29.** (C)	**30.** (B)
31. (C)	**32.** (B)	**33.** (B)	**34.** (C)	**35.** (B)	**36.** (B)	**37.** (B)	**38.** (B)	**39.** (A)	**40.** (B)
41. (C)	**42.** (C)	**43.** (B)	**44.** (C)	**45.** (B)	**46.** (B)	**47.** (D)	**48.** (A)	**49.** (D)	**50.** (B)